GP Model V3: Statistical Audit

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GP MODEL TECHNICAL AUDIT (V3) | idata_gp_v3.pkl
Configuration: 3 Chains (Full Run) | Matérn 3/2 Kernel
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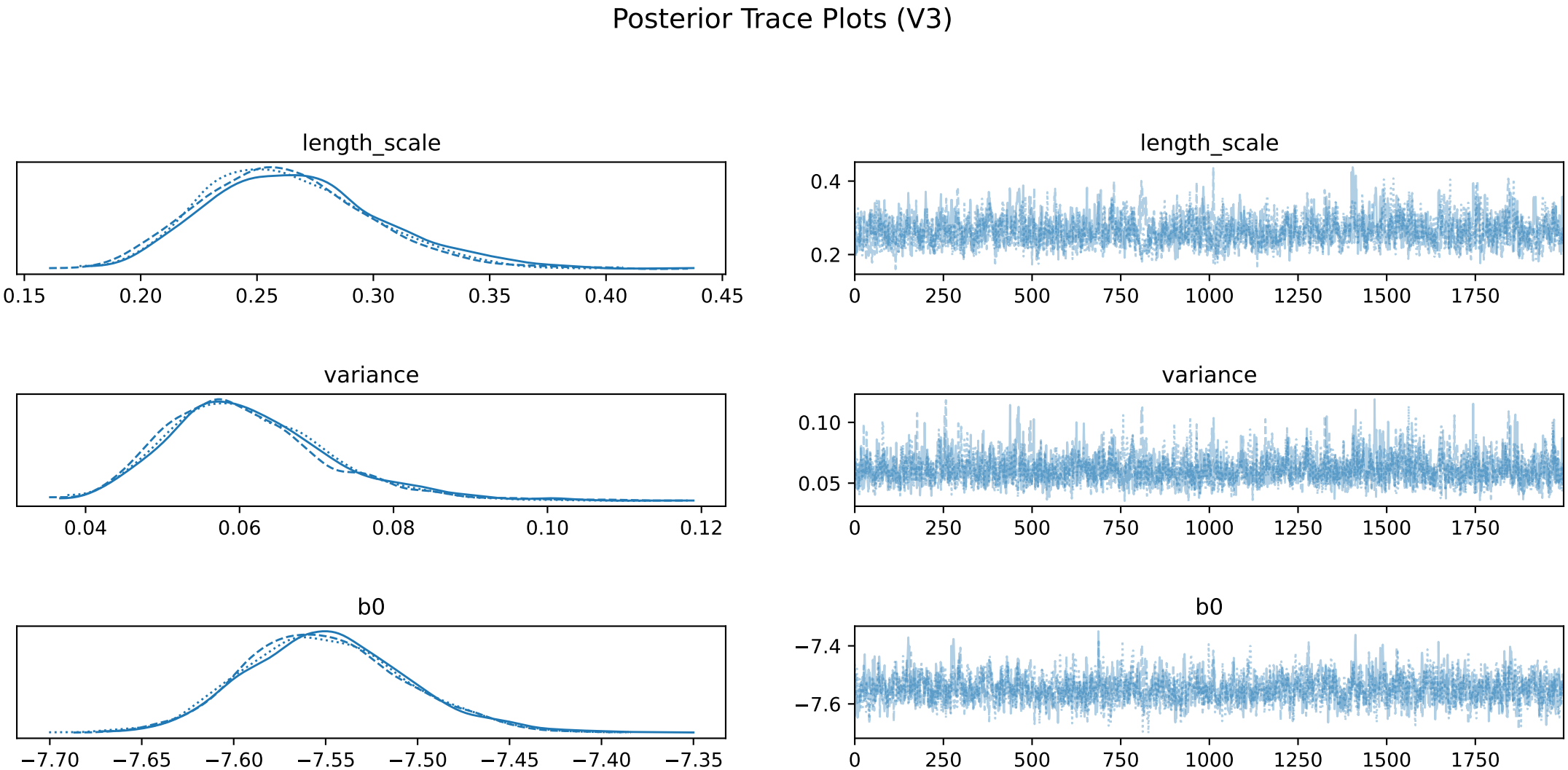
1. SAMPLING SPECIFICATIONS
------------------------------
Warmup Iterations:  2000
Sample Iterations:  2000
Total Post-Warmup:  6000

2. MCMC DIAGNOSTICS
------------------------------
Max R-hat:          1.000
Numerical Status:   STABLE

3. PARAMETER ESTIMATES
------------------------------
                mean     sd  hdi_3%  hdi_97%  r_hat  ess_bulk
length_scale  0.265  0.037   0.198    0.334    1.0     940.0
variance      0.061  0.011   0.042    0.082    1.0    1492.0
b0           -7.549  0.044  -7.627   -7.463    1.0    1409.0

4. SPATIAL CHARACTERISTICS
------------------------------
Mean Length Scale:  0.2650
Effective Range:    0.4590 (Standardized units)
Spatial Variance:   0.0610
Moran's I (Resid):  -0.1680
======================================================================
```


Posterior Trace Plots (V3)
length_scale
0.15
0.20
0.25
0.30
0.35
0.40
0.45
length_scale
0.4
0.2
0
250
500
750
1000
1250
1500
1750
variance
0.04
0.06
0.08
0.10
0.12
variance
0.10
0.05
0
250
500
750
1000
1250
1500
1750
b0
−7.70
−7.65
−7.60
−7.55
−7.50
−7.45
−7.40
−7.35
b0
−7.4
−7.6
0
250
500
750
1000
1250
1500
1750